1 Solution — GIAM 6.3 Problem 1

1.1 Problem

Consider the relation A defined by

$$A = \{(x, y) \ : \ x \text{ has the same astrological sign as } y\}.$$

Show that A is an equivalence relation. What equivalence class under A do you belong to?

1.2 Setup

Let P be a population of people, and let

$$\sigma : P \to Z$$

assign each person their zodiac sign, where $Z = \{\text{Aries}, \text{Taurus}, \ldots, \text{Pisces}\}$ has twelve elements. Then

$$A \ = \ \{(x, y) \in P \times P \ : \ \sigma(x) = \sigma(y)\}.$$

A relation R on a set X is an **equivalence relation** if it is reflexive, symmetric, and transitive.

1.3 A Is an Equivalence Relation

Each property of A follows from the corresponding property of equality on Z.

- **Reflexive.** For any $x \in P$, $\sigma(x) = \sigma(x)$, so $(x, x) \in A$.
- **Symmetric.** If $(x, y) \in A$, then $\sigma(x) = \sigma(y)$; equality on Z is symmetric, so $\sigma(y) = \sigma(x)$, hence $(y, x) \in A$.
- **Transitive.** If $(x, y), (y, z) \in A$, then $\sigma(x) = \sigma(y)$ and $\sigma(y) = \sigma(z)$; equality is transitive, so $\sigma(x) = \sigma(z)$, hence $(x, z) \in A$.

All three properties hold, so A is an equivalence relation. $\blacksquare$

This is the "pull back equality through a function" pattern: any relation of the form $\{(x, y) : f(x) = f(y)\}$ for $f : X \to Y$ is automatically an equivalence relation, with the three properties inherited from equality on Y.

1.4 Equivalence Classes

For each $x \in P$, the equivalence class of x under A is

$$[x]_A \;=\; \{y \in P \;:\; (x, y) \in A\} \;=\; \{y \in P \;:\; \sigma(y) = \sigma(x)\} \;=\; \sigma^{-1}\big(\sigma(x)\big).$$

So $[x]_A$ is the fiber of σ over $\sigma(x)$ — everyone in P sharing x's sign. Distinct signs give disjoint fibers, and the (at most) twelve non-empty fibers partition P:

$$P \;=\; \bigsqcup_{s \,\in\, \sigma(P)} \sigma^{-1}(s).$$

1.5 The Author's Class

The author's sign is **Scorpio**, so the author's equivalence class under A is

$$[\text{author}]_A \;=\; \sigma^{-1}(\text{Scorpio}) \;=\; \{\, y \in P \;:\; y \text{ is a Scorpio}\,\}.$$

Concretely, this is the set of every person in P born — in the Western tropical convention — between roughly 23 October and 21 November, placing the author alongside roughly $1/12$ of the population.

1.6 Remark — The Quotient Set

The **quotient set** P/A is the collection of all equivalence classes,

$$P/A \;=\; \big\{\, [x]_A \;:\; x \in P \,\big\}.$$

The map $x \mapsto [x]_A$ sends P onto P/A, and matches up with σ via $[x]_A \mapsto \sigma(x)$: each class corresponds to exactly one sign that actually occurs in P. Hence P/A is in bijection with the image $\sigma(P) \subseteq Z$, and

$$|P/A| \;=\; |\sigma(P)| \;\leq\; 12,$$

with equality whenever every sign appears in P — true for any reasonably large population.